



How Clockchain Compares with Other Timekeeping Systems

System	Primary Function	Accurate Time Reference	Tamper-Evident Record	Independently Verifiable Proof	Proof of Event Sequence
Clockchain	Creates independently verifiable evidence of time and event sequence	Yes — UTC-aligned	Yes	Yes	Yes — shared chronology
Network Time Protocol (NTP)	Synchronizes computer clocks over IP networks	Yes — implementation dependent	No	No	No
GPS time	Distributes a highly accurate global timing signal	Yes	No	No	No
Precision Time Protocol (PTP)	Provides high-precision synchronization within managed networks	Yes — implementation dependent	No	No	No
International Atomic Time (TAI)	Provides the continuous atomic timescale underlying UTC	Yes — not UTC itself	No	No	No
Blockchain block timestamps	Places transactions within a blockchain's sequence of blocks	Approximate	Yes	Yes — proof of inclusion	Yes — within one blockchain
Traditional audit logs	Record activity within an organization or system	System dependent	Sometimes	Usually no	Yes — within one log

NTP, GPS time, PTP and TAI provide or support accurate time, but they do not independently prove that a particular digital event occurred at the stated time or that its record has remained unchanged. Traditional audit logs preserve event histories, while blockchain block timestamps provide tamper-evident ordering within an individual blockchain; neither necessarily establishes a precise, externally verifiable time for events across different systems.

Clockchain combines a UTC-aligned time reference with decentralized validation and tamper-evident recording. This enables organizations to create independently verifiable evidence of when records were registered and the sequence in which recorded events occurred.

NOTE: The table describes the inherent or typical capabilities of each system category. Actual accuracy, security and auditability depend on implementation, configuration and supporting controls. TAI is an international atomic timescale rather than an event-timestamping or time-distribution system. Clockchain is intended to complement, rather than replace, existing synchronization and recordkeeping technologies.